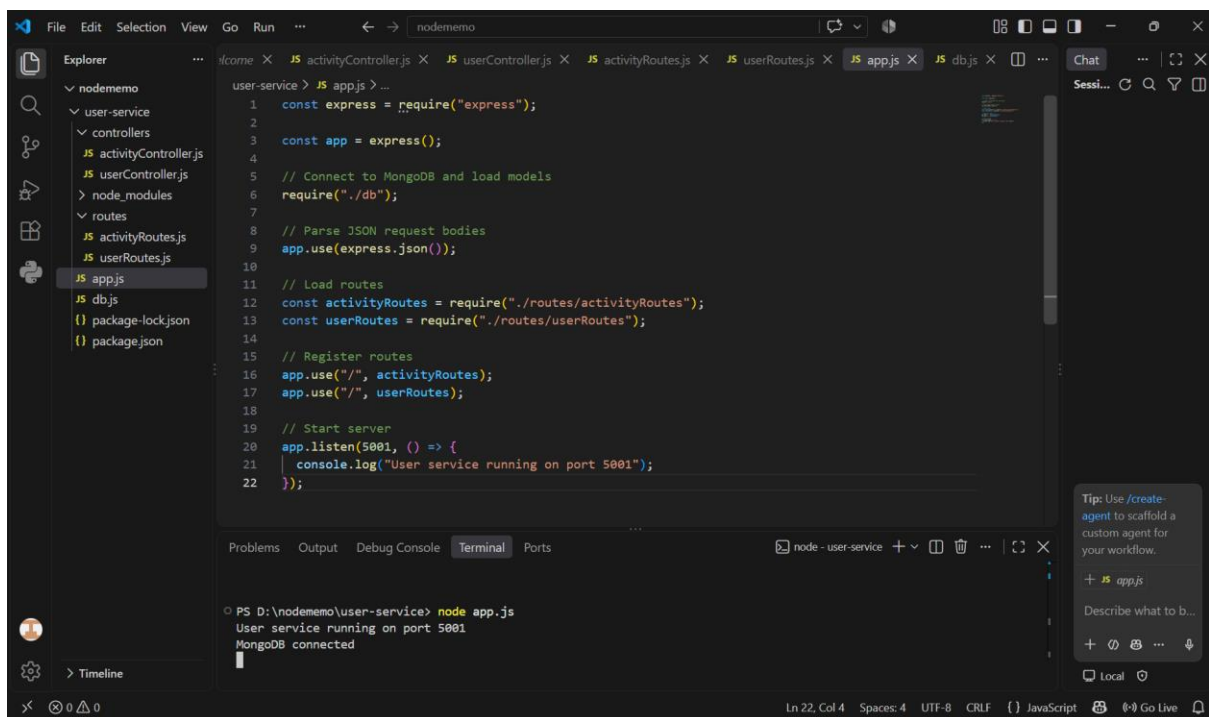


WEEK-7 PRACTICAL

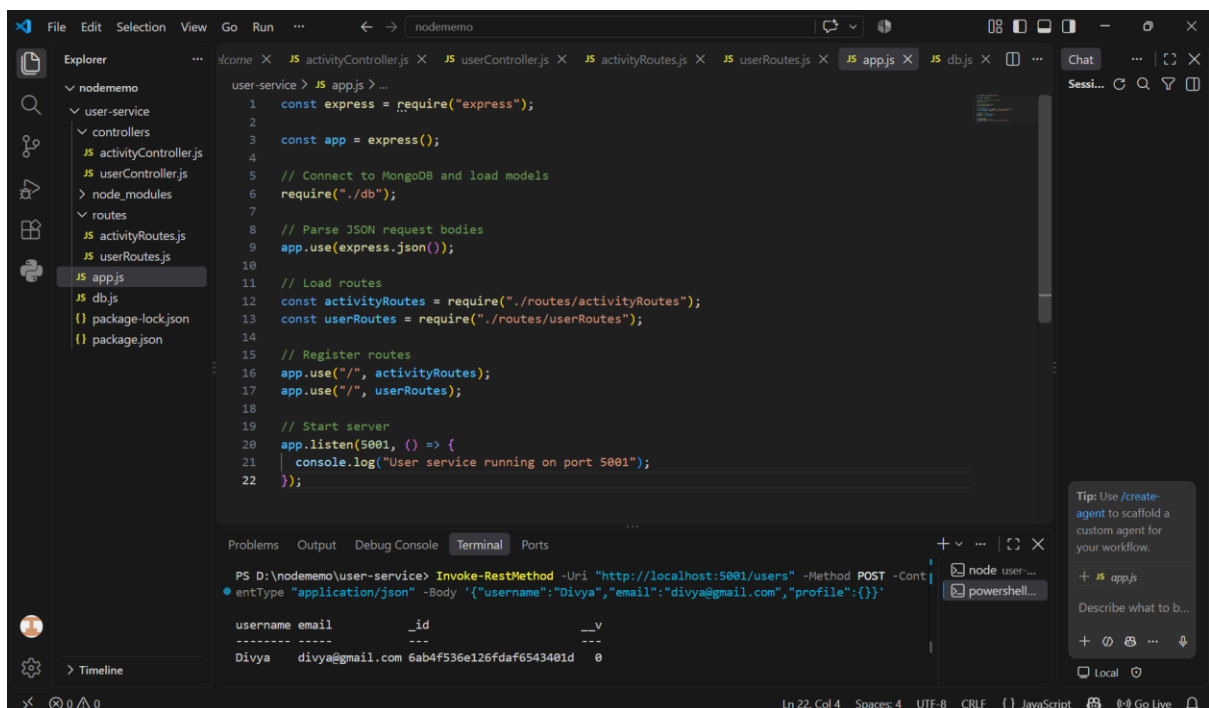


The screenshot shows the Visual Studio Code editor with a project named 'nodememo'. The Explorer sidebar on the left shows the file structure: 'user-service' contains 'controllers' (with 'activityController.js' and 'userController.js'), 'routes' (with 'activityRoutes.js' and 'userRoutes.js'), 'app.js', 'db.js', 'package-lock.json', and 'package.json'. The main editor displays 'app.js' with the following code:

```
1 const express = require("express");
2
3 const app = express();
4
5 // Connect to MongoDB and load models
6 require("./db");
7
8 // Parse JSON request bodies
9 app.use(express.json());
10
11 // Load routes
12 const activityRoutes = require("./routes/activityRoutes");
13 const userRoutes = require("./routes/userRoutes");
14
15 // Register routes
16 app.use("/", activityRoutes);
17 app.use("/", userRoutes);
18
19 // Start server
20 app.listen(5001, () => {
21   console.log("User service running on port 5001");
22 });
```

The Terminal at the bottom shows the command 'node app.js' being executed, with output: 'User service running on port 5001' and 'MongoDB connected'.

POST/USERS



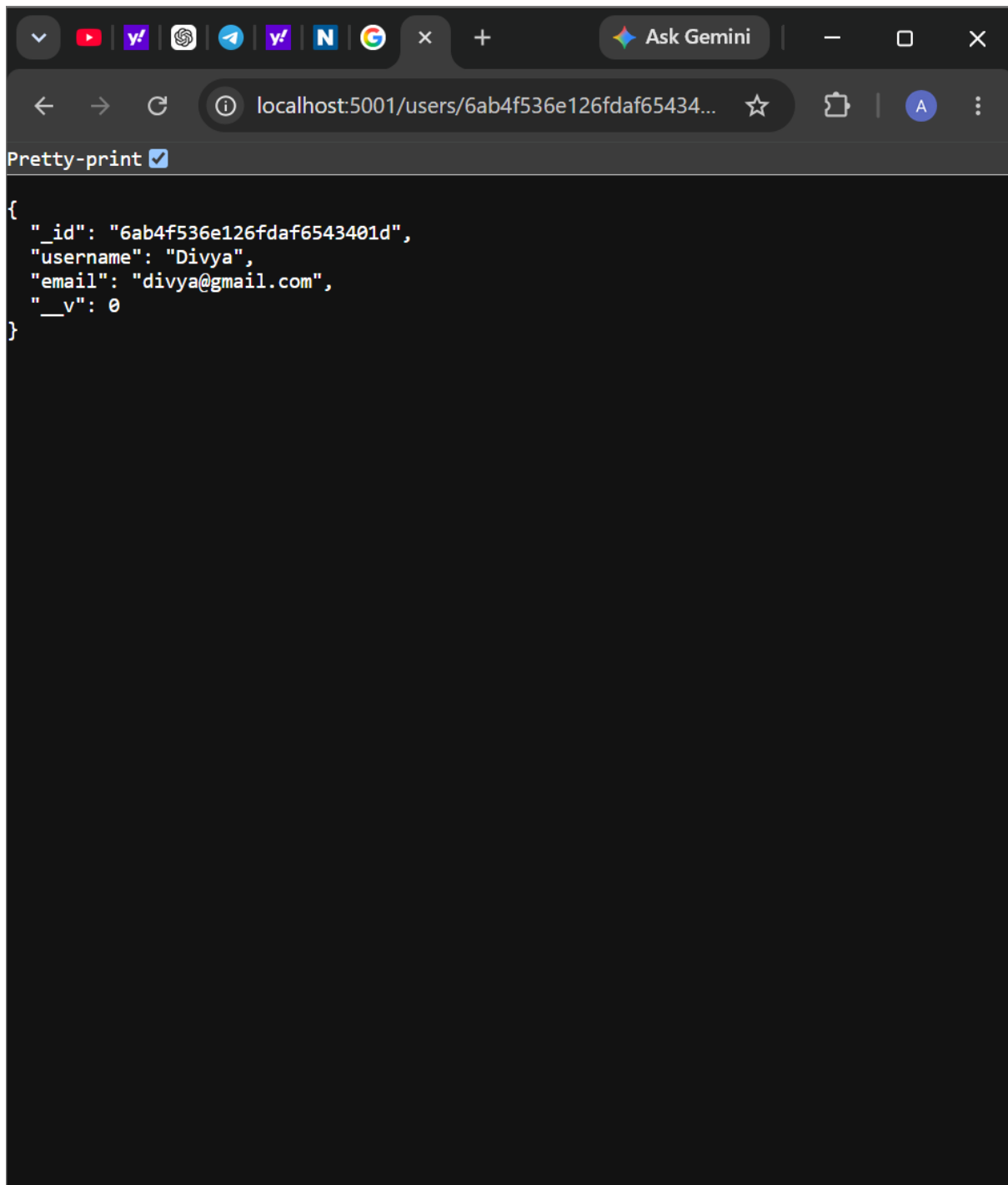
The screenshot shows the Visual Studio Code editor with the same project. The main editor displays 'app.js' with the same code as in the previous screenshot. The Terminal at the bottom shows a REST client request being executed:

```
PS D:\nodememo\user-service> Invoke-WebRequest -Uri "http://localhost:5001/users" -Method POST -Content '{"username":"Divya","email":"divya@gmail.com","profile":{}}'
```

The output of the request is displayed in a table:

username	email	_id	__v
Divya	divya@gmail.com	6ab4f536e126fdaf6543401d	0

GET/USERS



The image shows a web browser window with a REST client interface. The address bar displays the URL `localhost:5001/users/6ab4f536e126fdaf65434...`. The response body is a JSON object, which is formatted using the 'Pretty-print' feature. The JSON data represents a user with the following fields: `_id`, `username`, `email`, and `__v`.

```
{
  "_id": "6ab4f536e126fdaf6543401d",
  "username": "Divya",
  "email": "divya@gmail.com",
  "__v": 0
}
```

POST/ACTIVITIES

The screenshot shows the VS Code interface with the Explorer sidebar on the left displaying the project structure for 'nodememo' and 'user-service'. The main editor area shows the 'app.js' file with the following code:

```
1 const express = require("express");
2
3 const app = express();
4
```

The Terminal panel at the bottom shows two REST client requests being executed in a PowerShell environment:

Request 1:

```
PS D:\nodememo\user-service> Invoke-RestMethod -Uri "http://localhost:5001/users" -Method POST -ContentType "application/json" -Body '{"username":"Divya","email":"divya@gmail.com","profile":{}}'
```

Response 1:

```
user_id : 6ab4f536e126fdaf6543401d
action  : SEARCH_PYTHON
_id     : 6ab51c58e126fdaf6543401e
timestamp : 2026-09-24T12:49:28.283Z
__v     : 0
```

Request 2:

```
PS D:\nodememo\user-service> Invoke-RestMethod -Uri "http://localhost:5001/activities" -Method POST -ContentType "application/json" -Body '{"user_id":"6ab4f536e126fdaf6543401d","action":"VIEW_BOOK"}'
```

Response 2:

```
user_id : 6ab4f536e126fdaf6543401d
action  : VIEW_BOOK
_id     : 6ab51c98e126fdaf6543401f
timestamp : 2026-09-24T12:50:32.359Z
__v     : 0
```

The screenshot shows the VS Code interface with the Explorer sidebar on the left displaying the project structure for 'nodememo' and 'user-service'. The main editor area shows the 'app.js' file with the following code:

```
1 const express = require("express");
2
3 const app = express();
4
```

The Terminal panel at the bottom shows a REST client request being executed in a PowerShell environment:

```
PS D:\nodememo\user-service> Invoke-RestMethod -Uri "http://localhost:5001/activities" -Method POST -ContentType "application/json" -Body '{"user_id":"6ab4f536e126fdaf6543401d","action":"DOWNLOAD_BOOK"}'
```

Response:

```
user_id : 6ab4f536e126fdaf6543401d
action  : DOWNLOAD_BOOK
_id     : 6ab51d02e126fdaf65434020
timestamp : 2026-09-24T12:52:18.560Z
__v     : 0
```

GET/ACTIVITIES

The screenshot shows a web browser window with a dark theme. The address bar displays the URL `localhost:5001/activities/6ab4f536e126fdaf65...`. Below the address bar, a tab labeled "Pretty-print" is active. The main content area displays a JSON array of three activity records. Each record is an object with the following fields: `_id`, `user_id`, `action`, `timestamp`, and `__v`. The first record has an action of "DOWNLOAD_BOOK", the second has "VIEW_BOOK", and the third has "SEARCH_PYTHON".

```
[
  {
    "_id": "6ab51d02e126fdaf65434020",
    "user_id": "6ab4f536e126fdaf6543401d",
    "action": "DOWNLOAD_BOOK",
    "timestamp": "2026-09-24T12:52:18.560Z",
    "__v": 0
  },
  {
    "_id": "6ab51c98e126fdaf6543401f",
    "user_id": "6ab4f536e126fdaf6543401d",
    "action": "VIEW_BOOK",
    "timestamp": "2026-09-24T12:50:32.359Z",
    "__v": 0
  },
  {
    "_id": "6ab51c58e126fdaf6543401e",
    "user_id": "6ab4f536e126fdaf6543401d",
    "action": "SEARCH_PYTHON",
    "timestamp": "2026-09-24T12:49:28.283Z",
    "__v": 0
  }
]
```

MONGO DB:

MongoDB Compass - localhost:27017/digital_library

Connections Edit View Help

Compass

{ } My Queries

⚙ Data Modeling

CONNECTIONS (2)

Search connections

- csit2
- localhost:27017
 - admin
 - config
 - digital_library**
 - activities
 - users
 - local
 - studentDB
 - student_records

localhost:27017 > digital_library

Open MongoDB shell Create collection Refresh

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes
activities	-	32.77 kB	287.00 B	3	95.00 B	1
users	-	20.48 kB	78.00 B	1	78.00 B	1

MongoDB Compass - localhost:27017/digital_library.activities

Connections Edit View Collection Help

Compass

{ } My Queries

⚙ Data Modeling

CONNECTIONS (2)

Search connections

- csit2
- localhost:27017
 - admin
 - config
 - digital_library
 - activities**
 - users
 - local
 - studentDB
 - student_records

localhost:27017 > digital_library > activities

Open MongoDB shell

Documents 3 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or Generate query

Explain Reset Find Options

25 1 - 3 of 3

```
{ "_id": ObjectId('6ab51c58e126fdaf6543401e'),  
  "user_id": ObjectId('6ab4f536e126fdaf6543401d'),  
  "action": "SEARCH_PYTHON",  
  "timestamp": ISODate('2026-09-24T12:49:28.283+00:00')  
}
```

September 24, 2026 at 12:49:28 PM UTC

```
{ "_id": ObjectId('6ab51c98e126fdaf6543401f'),  
  "user_id": ObjectId('6ab4f536e126fdaf6543401d'),  
  "action": "VIEW_BOOK",  
  "timestamp": ISODate('2026-09-24T12:50:32.359+00:00')  
}
```

September 24, 2026 at 12:50:32 PM UTC

```
{ "_id": ObjectId('6ab51d02e126fdaf65434020'),  
  "user_id": ObjectId('6ab4f536e126fdaf6543401d'),  
  "action": "DOWNLOAD_BOOK",  
  "timestamp": ISODate('2026-09-24T12:52:18.560+00:00')  
}
```

September 24, 2026 at 12:52:18 PM UTC

MongoDB Compass - localhost:27017/digital_library.users

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

- csit2
- localhost:27017
 - admin
 - config
 - digital_library
 - activities
 - users
 - local
 - studentDB
 - student_records

Welcome student_records localhost:27017 users

localhost:27017 > digital_library > users

Documents 1 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

25 1-1 of 1

```
{
  "_id": ObjectId('6ab4f536e126fdaf6543401d')
  "username": "Divya"
  "email": "divya@gmail.com"
  "__v": 0
}
```